



Enhancement of the electrical performance of Ag/P3DDT/macroporous silicon Schottky diodes through thermal annealing process

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ABSTRACT

In this work, the impact of thermal annealing on the electrical performance of Ag/P3DDT/macroporous silicon (McPS) Schottky diodes was reported. The P3DDT polymer was deposited onto McPS by dip coating with controlled immersion time. McPS was previously prepared via silver-assisted chemical etching (MACE) method. Morphological and chemical analyzes revealed that immersion times between 30 and 60 min allowed for optimal polymer distribution, while longer times promoted excessive aggregation. Electrical characteristics were evaluated by current-voltage (I–V) measurements performed before and after annealing at different temperatures. The results show a significant improvement in diode performance with proper optimization of the immersion time and the annealing temperature. This improvement is manifested by higher rectification ratios, a reduced ideality factor of up to 1.15, and a decrease in series resistance. A reorganization of polymer chains on the porous matrix and the passivation of defects lead to improved carrier injection at the Ag-P3DDT and P3DDT-McPS interfaces, thus explaining the improved diode quality. This work offers new perspectives on organic/inorganic heterostructures, not only in the transport mechanism governing these interfaces but also in the crucial role of post-deposition heat treatment in modulating their electrical properties.

1. Introduction

The poly(3-alkylthiophene) (P3AT) polymer family has significant application potential in the field of organic electronics, due to its interesting optoelectronic properties [1–3]. P3ATs are conjugated thiophene ring-based polymers that exhibit good charge carrier mobility and strong π – π interaction between polymer chains [4,5]. Poly(3-dodecyl thiophene) (P3DDT), a P3AT polymer with semiconducting characteristics, plays an important role in the design and fabrication of electronic components [6]. In P3DDT structures, the low dielectric constant and the localization of electronic states strengthen the excitonic bond which are more stable and more influential than in inorganic materials, playing a central role in electronic transport mechanisms. In this context, electron spin plays a fundamental role in exciton dynamics. Indeed, singlet excitons are generally radiative while triplets have a much slower, often non-radiative recombination [6]. In other hand, macroporous silicon (McPS) morphology, which contains micrometer-

sized pores, gives it unique physical, electrical, and optical properties that differ significantly from bulk crystalline silicon. The formation of macropores or cavities within the silicon substrate introduces a high internal surface area and a network of depletion regions that significantly alter carrier transport compared to bulk Si. In McPS heterostructures, these effects manifest as modified Schottky barrier behavior, space-charge-limited currents, and often geometry-limited carrier mobilities, which are distinct from bulk silicon transport characteristics [7,8]. These structures have potential applications in several fields such as sensors, energy, and electronics [9–12]. McPS/Polymer composites possess a modified interface that adjusts the joining properties due to the large specific surface area of McPS and its different surface states. The organic materials such as conjugated polymers or small organic molecules are employed as the active layer in organic Schottky diodes (OSD) devices [13]. A variety of OSD based on silicon with various structures have been fabricated and studied [14–16]. Jiang et al. [17] reported that optimizing polymer infiltration in PEDOT:PSS/SiNW hybrid devices

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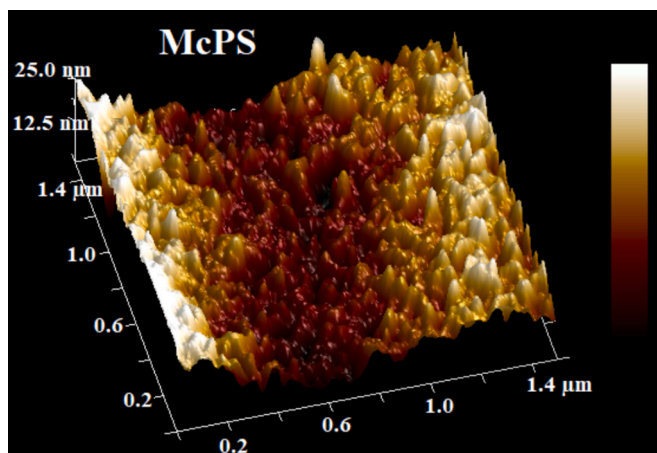


Fig. 1. 3D illustration of McPS area ($1.4 \mu\text{m} \times 1.4 \mu\text{m}$) scanned by AFM.

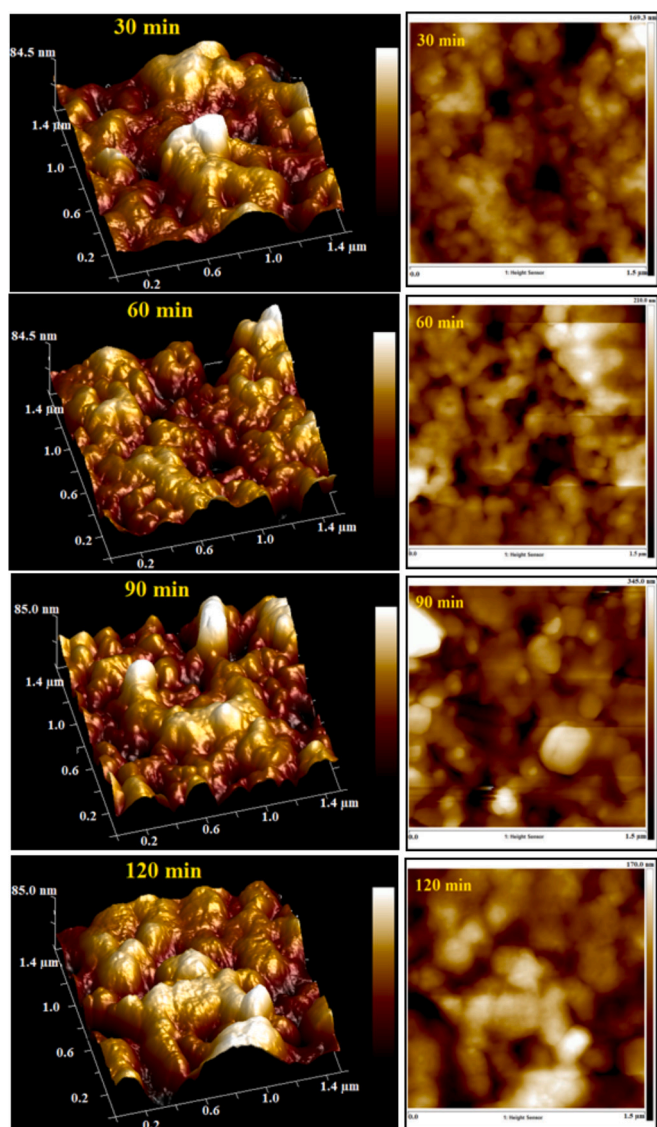


Fig. 2. 3D and 2D-AFM images of McPS/P3DDT samples for different immersion durations in P3DDT solution.

increased the power conversion efficiency of about 3%, accompanied by significant improvements in short-circuit current, open-circuit voltage,

and fill factor. These improvements were attributed to reduced carrier recombination, interfacial passivation, and more efficient charge collection at the organic/inorganic junction. Huanca [18] studied Schottky diodes formed by McPS and polyaniline polymer, he found a high ideality factor that varies between 5.8 and 37 and a potential barrier around 0.8 eV. In addition, the effect of high-temperature heat treatment on the properties of OSD diodes was an important factor for the application of this device [19,20]. High-temperature heat treatment can enhance electrical performance by improving contacts, reducing surface states, and enhancing crystallinity. This factor must be optimized to avoid degradation of organic layers or collapse of the porous structure. This is a critical step in tailoring the performance of OSD diodes to specific applications. Recently, Randrianantoandro et al. [21] demonstrated that Ag/TiO₂/macroporous silicon Schottky diodes exhibit performant electrical characteristics with optimal rectification behavior and improved diode parameters at specific conditions. The same study also revealed, through in-depth electrical measurements, the presence of pronounced interface effects as well as charge trapping phenomena, highlighting the strong potential of macroporous silicon functionalized with TiO₂ for optimizing the performance of electronic devices. In this paper, we present an electrical study by I-V measurements of P3DDT deposited on McPS surface with emphasis on the effect of annealing temperature on the electrical properties of Ag/P3DDT/McPS/Si diodes. We investigated the combined effects of P3DDT immersion time and post-deposition thermal annealing on interface quality, morphological organization and charge transport, providing new perspectives for the optimization of organic/inorganic heterojunctions. We note that the originality of this work lies mainly in: (i) the correlated morphological and electrical analysis linking the distribution of P3DDT, the evolution of roughness, and the transport regimes, (ii) the in-depth study of the impact of annealing on the Schottky barrier parameters, and (iii) the identification of optimal processing conditions that minimize the series resistance and the ideality factor of Ag/P3DDT/McPS diodes.

2. Experimental

The substrate was a p-type Si (100) with 0.1–2 $\Omega \text{ cm}$ resistivity. Macroporous silicon (McPS) was prepared by Ag-Assisted chemical etching process (Ag-ACE). Cleaning with deionized water and acetone for 30 min is necessary to remove organic residues on the surface of the substrates. Silicon etching was performed using a solution composed of HF (5 M) and AgNO₃ (0.05 M) in a 1:1 volume ratio for 20 min. Following etching, the samples were treated with nitric acid for 5 min, rinsed with deionized water, and dried at room temperature to remove the silver dendrites formed as a result of the chemical reduction of Ag⁺ ions. A polymeric solution with a concentration of 5 mg/mL was prepared by dissolving rr-P3DDT powder (with a regioregularity greater than 98%) in toluene. The deposition of P3DDT onto the macroporous silicon (McPS) surface was carried out by electroless deposition without stirring to prevent turbulence during the reaction. An ohmic contact of 200 nm thick aluminum was deposited on the backside of the silicon wafer using the evaporation method. Ag/P3DDT/McPS diodes were fabricated by applying conductive silver paste for the top contacts, which were circular (approximately 5 mm in diameter and 0.5 mm thick), and cured at 60 °C for 30 min. All chemicals used in the Ag-assisted chemical etching (Ag-ACE) process, including HF, AgNO₃, and silver paste, were purchased from Sigma-Aldrich. Current-voltage (I-V) measurements of the developed junctions were performed over a voltage range of −10 V to +10 V using a synchronous detector (SR830), controlled via LabVIEW software. These measurements were conducted at room temperature under dark conditions. Thermal annealing was carried out in air for temperatures ranging from 50 °C to 200 °C with a step of 30 °C using a programmable furnace, with a 15-min stabilization period at each temperature step. A FTIR spectrometer of Bruker IFS66v/s type was used for performed FTIR analyses. FTIR spectra are recorded on

Table 1

Mean surface roughness (Ra and RMS) of McPS/P3DDT samples for different immersion durations in P3DDT solution.

$t_{\text{imm}}(\text{min})$	0	30	60	90	120
RMS (nm)	8.5	12	15.5	22.1	28.2
Ra (nm)	6.8	9.6	12.2	17.6	22.6

transmittance mode in the 400–3500 cm^{-1} range with a step of 2 cm^{-1} . The SEM images were obtained using a field-emission scanning electron microscope of SU-8030 type from Hitachi High-Tech group. The AFM observations were performed by an Atomic Force Microscope (AFM, BRUKER, Dimension ICON in “tscan asyst” mode or intermittent contact).

3. Results and discussion

3.1. Morphological properties

Fig. 1 displays the AFM image of McPS before immersion in the P3DDT polymer solution. This image shows a highly textured surface, characteristic of porous structures obtained by Ag-assisted chemical etching. The height variations suggest significant surface porosity, with no apparent presence of covering materials.

Fig. 2 presents the AFM images of McPS impregnated in the P3DDT

solution for durations of 30, 60, 90 and 120 min.

The AFM image reveals a relatively rough surface, with heights up to ~25 nm, suggesting inhomogeneous deposition of the P3DDT polymer on the macroporous silicon surface. The wavy and granular morphology may indicate the formation of polymer domains, possibly related to aggregation or partial infiltration into the pores. The underlying structure of the McPS appears to be partially masked by the P3DDT molecules. More pronounced reliefs and more bumps when the immersion time changes from 60 min to 120 min. Some peaks appear more pronounced than in the first image, and the surface exhibits a more granular morphology, suggesting an accumulation of P3DDT. The surface of the McPS seems more covered by the polymer. By further increasing the immersion time, we observe a much more chaotic morphology, with areas of high height (white tips). The P3DDT seems to have formed clusters, or completely covered the underlying structures. We observe a decrease in homogeneity, probably linked to aggregates or saturation of the surface. Increasing the immersion time of the McPS in the P3DDT solution leads to a progressive evolution of the surface morphology, characterized by an increase in roughness, an increasingly complete coverage of the porous structure, and, for the longest times, the formation of polymer clusters. P3DDT first adsorbs on the inner walls of McPS and then accumulates on the surface over time. Excessive immersion time (> 60 min) can lead to aggregation, reducing the quality of the polymer/McPS interface. The evolution of morphologies can affect the electronic properties and the conduction mechanism.

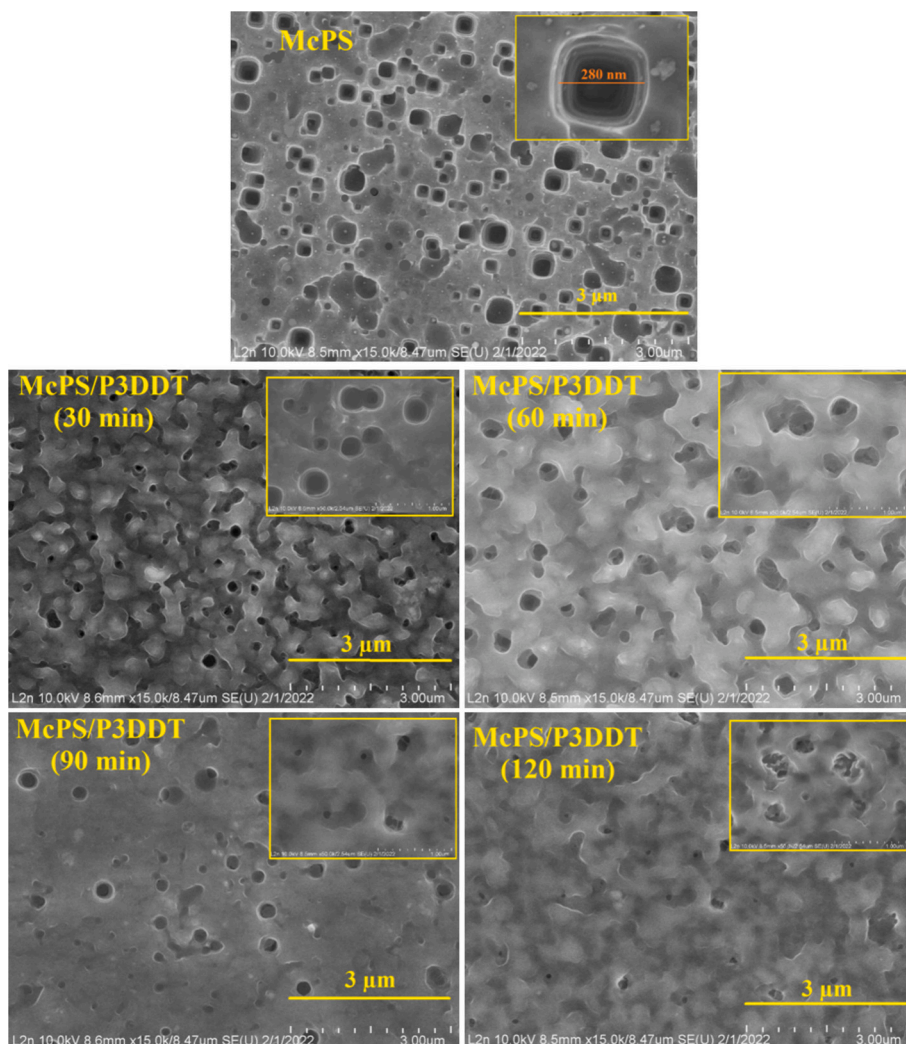


Fig. 3. Plan view SEM images of McPS and McPS/P3DDT samples for different immersion durations in P3DDT solution.

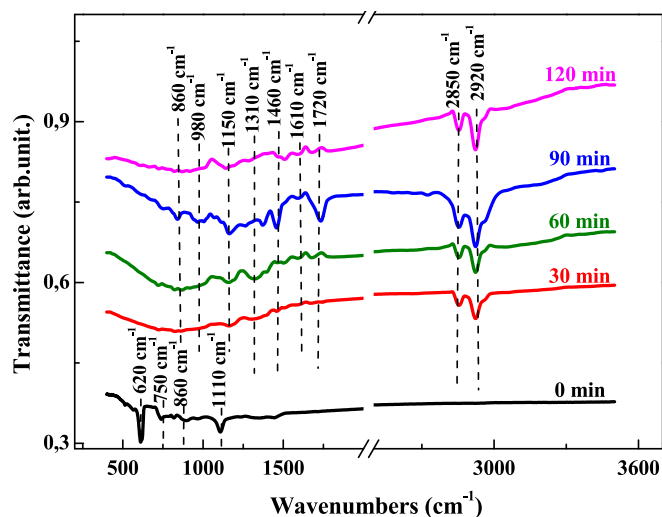


Fig. 4. Influence of immersion time on the FTIR spectrum of McPS coated with P3DDT.

A monotonic and significant increase in both roughness parameters is observed with immersion time (Table 1). The RMS value increases from 8.5 nm before immersion to 28.2 nm after an impregnation time of 120 min in the P3DDT solution. This strong growth reflects the appearance of increasingly pronounced surface features and high peak heights, corresponding to polymer aggregates. The roughness R_a follows a similar trend, increasing from 6.8 nm to 22.6 nm. R_a is a key parameter for evaluating the effectiveness of surface coating. A high R_a value indicates more pronounced surface roughness, associated with the presence of peaks and pits. This quantitative increase in roughness is the direct signature of P3DDT accumulation on the McPS surface. The process is not simply a monolayer formation, but rather a three-dimensional growth that leads to an increasingly rough topography [22].

Fig. 3 illustrate SEM images of the surface of bare macroporous silicon (McPS) and coated with P3DDT for different immersion times (30, 60, 90 and 120 min).

The surface of the McPS before immersion exhibits a well-defined

macropore distribution. The pores are nearly circular to polygonal in shape, with an average diameter of approximately 280 nm, as shown in the enlarged inset image. The pore walls are sharp and well-contrasted, indicating a clean, uncoated silicon surface. This morphology is typical of McPS obtained MACE method [23], providing a large specific surface area that constitutes an excellent matrix for P3DDT infiltration.

For 30 min, P3DDT begins to deposit preferentially on the pore walls and between the cavities. The pores remain mostly open, but their apparent diameter decreases slightly. As the immersion time increases, deeper infiltration into the pores is observed, along with polymer coalescence and a transition to a more compact McPS/P3DDT hybrid layer. For 60 and 90 min, the visible pores are significantly fewer and smaller in size. The polymer acts as a filler material; the porous surface is almost completely covered by the P3DDT polymer, thus confirming the AFM observations. For 120 min, the morphology is dominated by a nearly compact polymer layer. The pores are largely clogged or completely masked. The surface exhibits a granular texture, and the underlying porous substrate is almost indistinguishable. Therefore, the observed morphological evolution suggests a three-step mechanism. An initial adsorption of P3DDT on the internal walls of the McPS (30 min), then an infiltration with progressive coating of the pores (60–90 min) before clogging and formation of a continuous polymer film on the surface for 120 min.

The morphological nature of McPS facilitates the infiltration of P3DDT molecules into the pores. Conversely, in matrices with finer porosity, such as silicon nanowires and nanoporous silicon, polymer infiltration, due to their macromolecular nature, remains highly limited [24,25]. The progressive evolution of the morphology and degree of infiltration of P3DDT into the macroporous structure of silicon is likely to significantly influence the charge transport mechanisms and electrical parameters of the resulting diodes. The study of this impact is the subject of the following section.

3.2. Chemical composition

FTIR spectroscopy was carried out in order to determine the chemical composition changes that can take place on the MPS surface after the deposition of the polymer. The FTIR spectra of McPS and McPS/P3DDT nanocomposites elaborated at immersion durations of 30 min, 60 min, 90 min and 120 min are shown in Fig. 4. The FTIR spectrum of

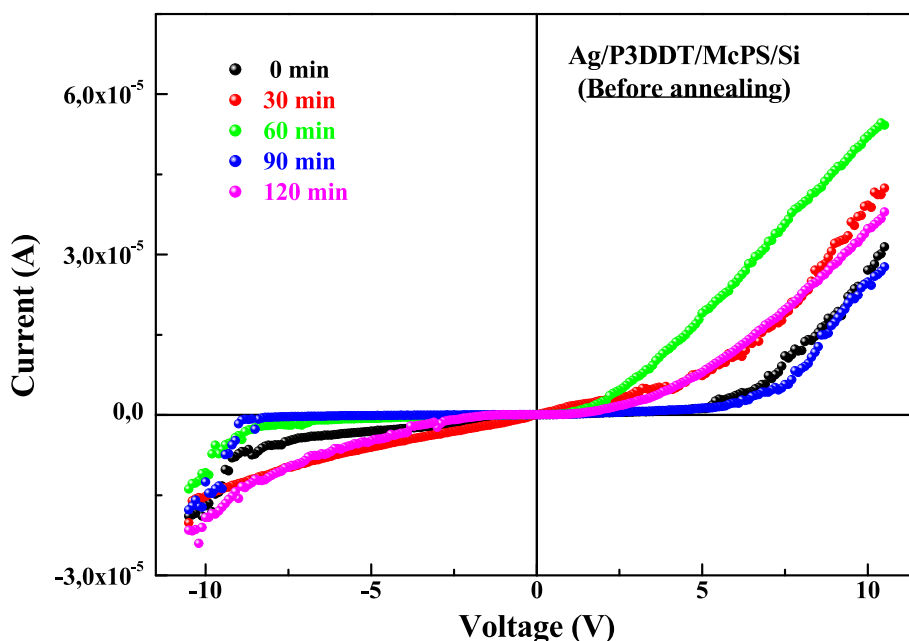


Fig. 5. Immersion-time effects on the I–V profile of Ag/P3DDT/McPS diodes.

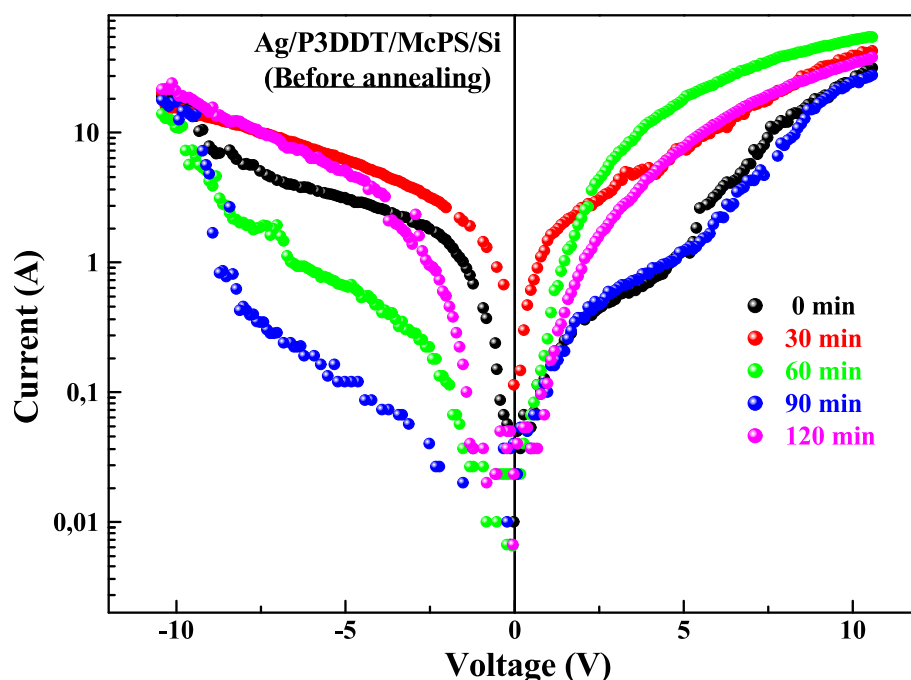


Fig. 6. Effects of immersion time on the semi-logarithmic I–V characterization of Ag/P3DDT/McPS heterojunctions.

the McPS sample before impregnation (0 min) closely matches those reported in previous studies [26,27]. The most prominent absorption bands appear at 1110 cm^{-1} and 620 cm^{-1} , which are attributed to the stretching vibrations of Si–O–Si and the bending vibrations of Si–H_n ($n=1, 2$), respectively.

The FTIR band at 1150 cm^{-1} corresponds to the vibrations of the Si–O–Si bonds [28]. New peaks appear in the spectra of McPS immersed in P3DDT and are clearly distinguished for duration of 90 min. These peaks, located at 980 , 1460 , 1610 and 1720 cm^{-1} , are attributed to Vinyl terminals, CH₃ (Bending) or aromatic –C=C– (Stretching), C=C (Stretching), and C=O (Stretching), respectively [29–32]. The increasing intensity of these FTIR bands with an immersion time of 90 min confirms that the deposition of P3DDT on the McPS is significant. This indicates a good interaction between the polymer and the substrate and a better molecular organization with the immersion time. The intensity of the different vibration bands weakens for immersion duration of 120 min indicating weak interfacial interactions, thus marking a saturation of the McPS surface by the polymer molecules. Intense peaks were observed at 2850 and 2920 cm^{-1} which are characteristic of CH₂ (Stretching) and CH₂ (Antisymmetric stretching) vibrations, respectively [33].

After impregnation of McPS in the P3DDT solution, the Si–H_n bonded band disappears and is replaced by new absorption bands associated with the polymer. This indicates the substitution of surface hydrogen bonds by polymer groups, particularly through the formation of Si–C(H) bonds. This FTIR study shows the strong chemical reactivity between the surface of McPS and the molecules of the P3DDT polymer.

3.3. Electrical characterizations

The electrical investigations of Ag/P3DDT/McPS/Si junctions were performed through current–voltage (I–V) measurements. This study allows us to analyze the diodes parameters as well as the charge transport behavior as a function of immersion durations, both before and after annealing at temperatures ranging from $50\text{ }^{\circ}\text{C}$ to $200\text{ }^{\circ}\text{C}$.

3.3.1. Electrical study before annealing treatment

Fig. 5 displays the I–V characteristics of Ag/P3DDT/McPS/Si

junctions for immersion times ranging from 0 to 120 min before any thermal annealing.

All curves exhibit nonlinearity typical of semiconductor junctions, suggesting Schottky diode-like behavior. The current increases with immersion time up to 60 min, then decreases for 90 min, and remains relatively high for 120 min. The different diodes exhibit asymmetric I–V characteristics. The threshold voltage is higher with the presence of the P3DDT polymer. Generally, the turn-on voltage is the forward voltage at which the diode starts conducting significantly. For organic Schottky diode (OSD), this voltage is often higher than in inorganic counterparts due to the lower mobility of charge carriers in organic materials.

Fig. 6 shows the semi-log I–V characteristics of the Ag/P3DDT/McPS Schottky diode for different immersion durations. Such a representation allows comparing the I–V characteristics even for areas where the currents are very close at low applied voltages.

The higher current for 60 min indicates optimal thickness or coverage of the P3DDT polymer on the McPS, promoting charge transport. For shorter times (30 min), deposition may be incomplete, limiting charge transport. For longer times, a large or disordered accumulation of P3DDT molecules on the McPS surface could introduce charge traps, thus reducing the current. A rectifying character is clearly observed for average immersion times (60 min and 90 min). A small current variation was observed at voltages of -10 V and $+10\text{ V}$. This is mainly attributed to charge transport limited by the interface at the Ag/P3DDT/McPS junction, where the applied electric field is insufficient to significantly reduce the Schottky barrier or activate field-enhanced transport [34,35]. Although the current response is almost the same for high applied voltages, a clearer distinction appears at low forward bias ($< 5\text{ V}$), particularly for the 60 and 90 min samples, where transport is strongly influenced by interface states and trap-assisted jumping [36].

Recall that AFM and SEM analyses confirm that longer deposition times increase surface roughness and polymer aggregation. This morphology favors trap-based capture and release processes, consistent with interfacial leakage models involving barrier-controlled Schottky emission. These effects modify the low-polarization response while maintaining an interface-dominated conduction regime.

We present in Fig. 7 the I–V characteristics corresponding to the Ag/P3DDT/McPS diodes for immersion times varying from 30 to 120 min,

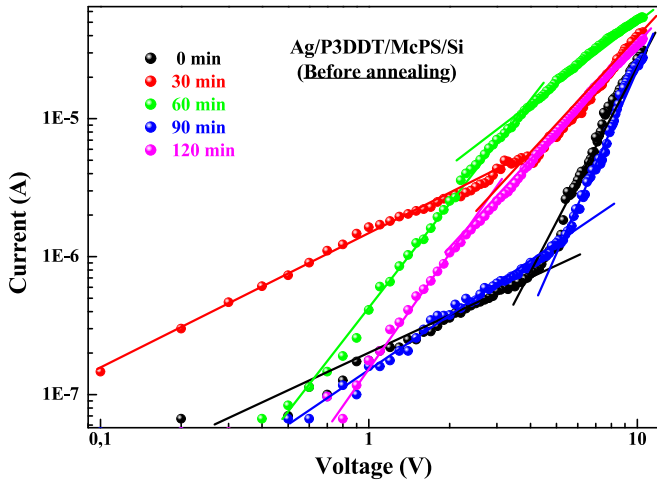


Fig. 7. The logarithmic I-V characteristics of the Ag/P3DDT/McPS Schottky diode for different immersion durations.

Table 2

The values of m obtained at low voltage (L. V.) and high voltage (H. V.) from the logarithmic I-V characteristics for different immersion times.

t_{imm} (min)	0	30	60	90	120
L. V.	1.08	0.91	2.51	1.34	2.37
H. V.	4.19	2.52	1.5	4.68	2.09

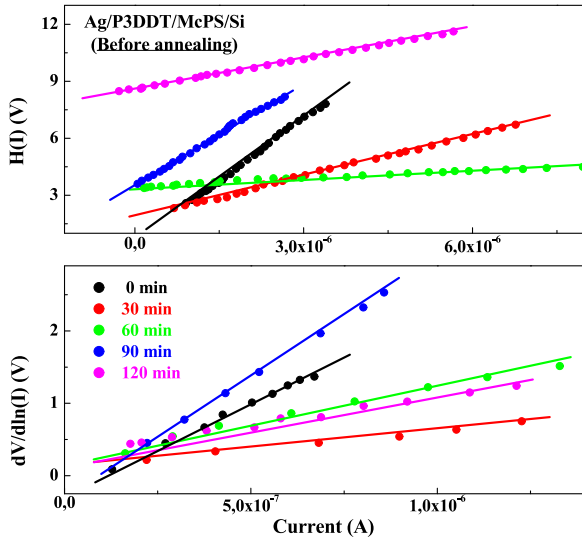


Fig. 8. $dV/d\ln(I)$ and $H(I)$ plots of the Ag/P3DDT/McPS Schottky diode for different immersion durations.

Table 3

Diode parameters of Ag/P3DDT/McPS diode for different immersion times before annealing treatment.

t_{imm} (min)	0	30	60	90	120
n	8.64	4.81	9.58	10.23	11.37
R_s (k Ω)	2560	701	640	1035	531
Φ_b (eV)	0.634	0.397	0.536	0.693	0.778

plotted on a logarithmic scale.

The identification of the different conduction regimes is primarily based on the I-V power-law behavior, which is a commonly adopted and

well-established approach in the literature for similar organic/porous silicon diode structures [37]. A linear variation is observed, accompanied by a notable change in slope when transitioning from low to high polarization regions indicating a power-law behavior ($I \propto V^m$), where the slope gives the exponent m . The different values of m are given in Table 2.

At low voltages, the ohmic regime is observed for immersion times of 0 min and 30 min where ' m ' remains close to 1. Such behavior suggests which suggests ohmic conduction or linear transport, where the diode behaves like a resistor. For 60 min and 120 min, the current is space charge limited (SCLC) at low applied voltages [38]. This mechanism occurs when the injected charge carriers exceed the free carrier concentration in the semiconductor, creating a space charge field that limits the current, according to the Child-Langmuir law [39]. At high voltages, the very large slopes ($m \approx 4$ for 0 and 90 min) indicate the trap-filled limit (TFL) regime, where most traps are saturated and the current increases sharply with voltage [40]. This behavior is typical of organic semiconductors with high trap densities or disordered morphology [19,20]. Conversely, the lower slopes for the 60 min suggest a more homogeneous polymer layer with fewer traps, enabling quasi-linear or moderate SCLC conduction. The strong dependence of m on immersion time reflects the influence of P3DDT distribution and interface quality. In fact, short or excessive deposition times lead to poor coverage or aggregation, increasing trap density and promoting TFL behavior, while optimized deposition (60 min) minimizes these effects.

Given that the P3DDT polymer and McPS exhibit the behavior of lightly doped p-type materials, the electrical characteristics of the different diodes were analyzed using thermionic emission theory [41]. Based on this theory, and considering the series resistance as a corrective term, the current as a function of voltage is given by:

$$I = I_s \exp \left[\frac{q(V - R_s I)}{nKT} \right] \quad (1)$$

where V is the applied voltage, q is the elementary charge, T is the absolute temperature, k is Boltzmann's constant, I_s is the saturation current, and n is the ideality factor.

The diode parameters were extracted using the method proposed by Cheung and Cheung [42]. The corresponding functions, derived from Eq. (1), are expressed as follows:

$$\frac{dV}{d\ln(I)} = IR_s + n \frac{kT}{q} \quad (2)$$

$$H(I) = n \left(\frac{kT}{q} \right) \ln \left(\frac{I}{AA^* T^2} \right) \quad (3)$$

$$H(I) = IR_s + n\phi_b \quad (4)$$

A , A^* , and ϕ_b are the diode area, the effective Richardson constant, and the energy barrier, respectively. Fig. 8 shows the $dV/d\ln(I)$ and $H(I)$ plots of the Ag/P3DDT/McPS Schottky diode for different immersion durations. The diode parameters are highly dependent on the selected voltage range. Therefore, the Cheung model must be applied within the appropriate voltage region, excluding very low voltages dominated by recombination and trapping effects, and high voltages affected by series resistance.

The results obtained by fitting the I-V characteristics with the two Cheung equations are listed in Table 3.

The ideality factor is high for all samples, indicating non-ideal diode behavior and suggesting transport mechanisms other than simple diffusion involving inhomogeneous barrier effects as well as defect-induced traps. This behavior suggests that additional transport mechanisms, such as interface states, barrier inhomogeneity, and the presence of the intermediate polymer layer, continue to dominate charge transport. Moreover, for the structure without polymer (0 min), the large specific surface area induced by the macroporous silicon promotes large

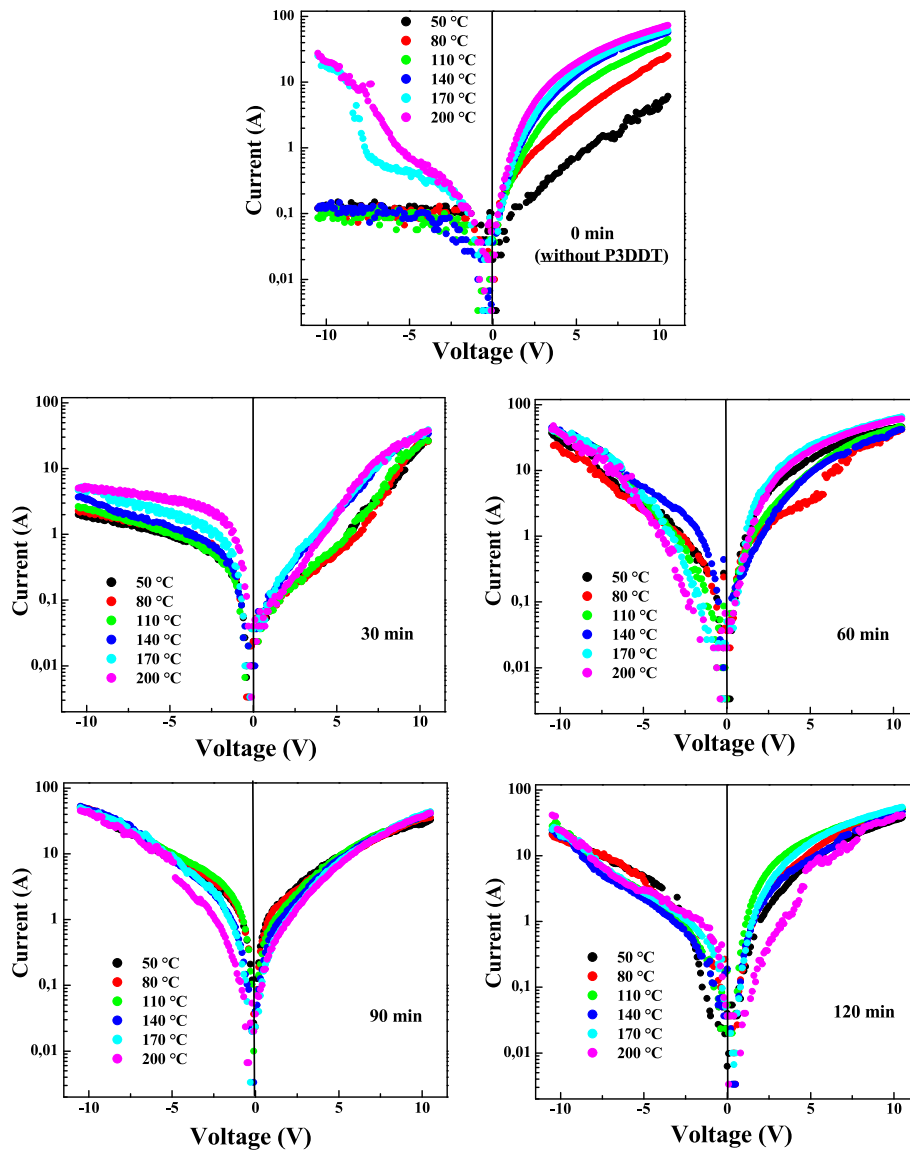


Fig. 9. The semi- log I-V characteristics of the Ag/P3DDT/McPS Schottky diodes for different immersion durations as a function of annealing temperature ranging from 50 °C to 200 °C.

Table 4

The m values obtained from the $\log(I)$ – $\log(V)$ characteristics at low and high voltage regimes for different immersion times and annealing temperatures.

	A.T.	50 °C	80 °C	110 °C	140 °C	170 °C	200 °C
$t_{\text{imm}} = 0 \text{ min}$	L. V.	1.21	1.2	1.99	2.05	2.21	2.12
	H. V.	2.93	2.8	2.27	2.03	2.2	1.78
	V.						
$t_{\text{imm}} = 30 \text{ min}$	L. V.	1.21	1.07	1.19	1.44	1.6	1.36
	H. V.	5.29	7.14	5.86	4.37	4	4.34
	V.						
$t_{\text{imm}} = 60 \text{ min}$	L. V.	1.99	1.18	1.78	1.4	2.94	2.6
	H. V.	1.59	3.49	2.19	2.23	1.45	1.44
	V.						
$t_{\text{imm}} = 90 \text{ min}$	L. V.	0.96	0.94	1.08	1.26	1.43	1.4
	H. V.	1.7	2.07	2	2.19	2.39	2.47
	V.						
$t_{\text{imm}} = 120 \text{ min}$	L. V.	2.45	2.53	2.89	2.4	3.07	2.93
	H. V.	2.06	2.06	1.43	2.18	1.78	3.1
	V.						

native oxidation inside the pores. This results in increased series resistance and pronounced barrier inhomogeneity, leading to ideality factors

significantly higher than unity. Such behavior is well documented in porous silicon-based Schottky diodes and is commonly attributed to native oxide layers and interface states [7,43]. We note that a high ideality factor is frequently observed in Schottky diodes incorporating an organic interlayer since the organic/metal or organic/semiconductor interface may have irregular surface states or barriers [44,45]. The organic layer may also generate traps that slow or deflect carrier transport, thus increasing non-ideality. The n value decreases sharply at 30 min and then rises to even higher values for the other times, indicating better junction quality at 30 min. At 60, 90, and 120 min, the high values of n suggest a large change in interface quality due to excessive polymer accumulation, thus influencing conduction. We note that disordered morphology at the P3DDT/McPS interface can introduce localized defect states that act as charge traps, significantly affecting charge transport and the electrical characteristics of diodes [46–48].

The series resistance decreases sharply up to 60 min ($R_s = 640 \text{ k}\Omega$), which reflects an improvement in conduction indicating a better electrical contact which leads to a reduction of interfacial defects and an improvement of the P3DDT layer. But at 90 min, it rises sharply to 1035 $\text{k}\Omega$, probably a sign of excessive thickening of the polymer or poor infiltration into the pores of the McPS. Regarding the potential barrier

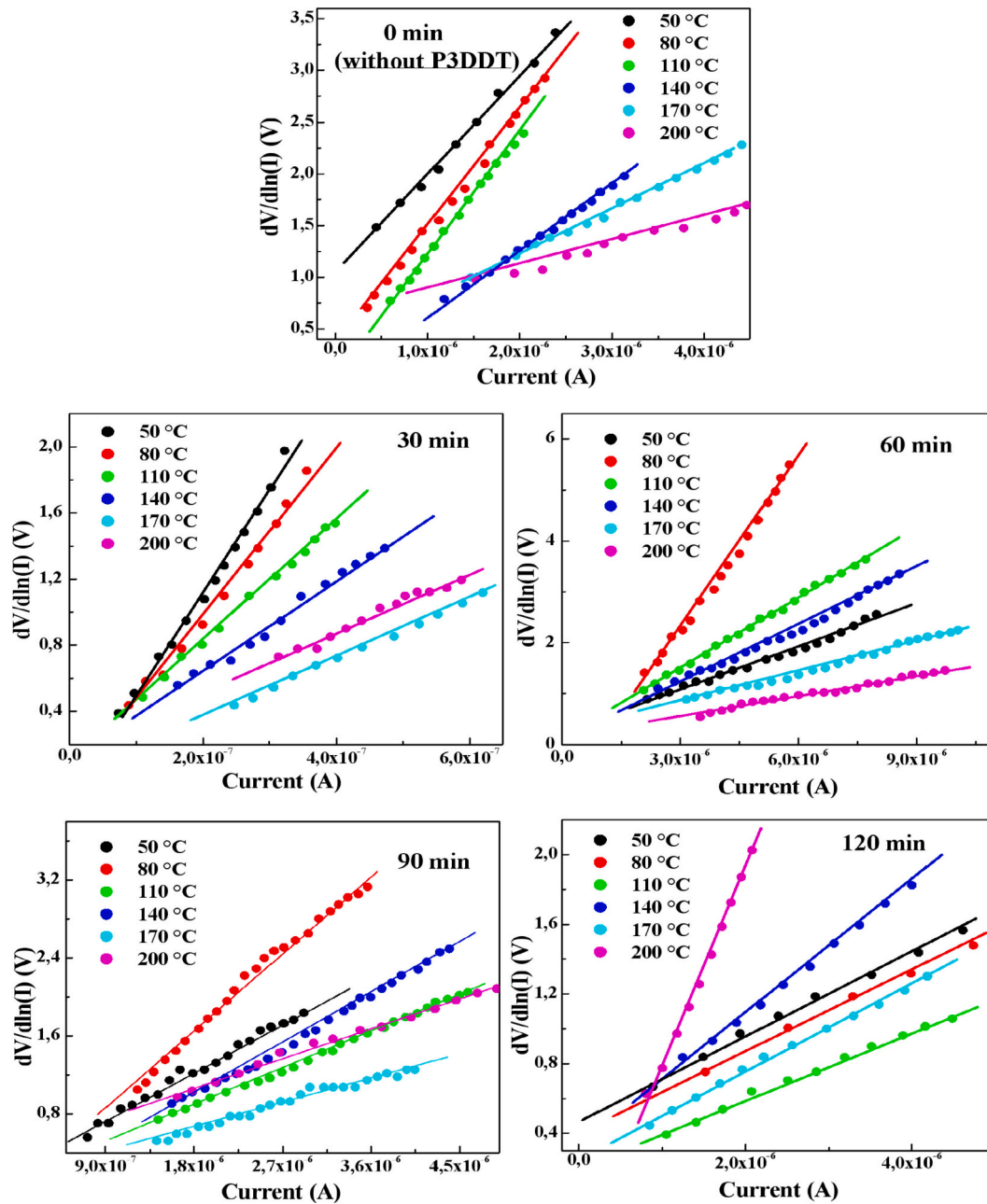


Fig. 10. The $dV/d\ln(I)$ plots of the Ag/P3DDT/McPS Schottky diodes for different immersion durations as a function of annealing temperature ranging from 50 °C to 200 °C.

(Φ_b), it increases overall with immersion time, which is favorable for junction formation. It increases from 0.397 eV to 0.778 eV, indicating better interface quality and reduced interfacial defects with increasing immersion time. However, the fact that the barrier height increases while the ideality factor remains high (or increases) for immersion times superior to 60 min indicates that there are trapping effects or inhomogeneities despite the good barrier.

3.3.2. Electrical study after annealing treatment

The semi-log I-V characteristics of the Ag/P3DDT/McPS Schottky diodes for different immersion durations as a function of annealing temperature ranging from 50 °C to 200 °C are shown in Fig. 9.

From this figure, the I-V behaviors are typical of Schottky diodes, with an asymmetry between the forward and reverse polarities, although this asymmetry varies greatly depending on the processing temperatures.

The forward bias current increases with temperature up to a certain limit (140 °C - 170 °C). Moderate annealing improves charge transport by reducing defects or improving interchain conduction of the polymer. For an immersion time of 120 min, the current decreases in forward bias when the annealing temperature reaches 200 °C, compared to other temperatures under the same measurement conditions. This could be due to thermal degradation of the polymer or unwanted chemical reactions. The linearity of the $\log(V)$ vs $\log(I)$ variation still persists for all annealing temperatures with a change in slope by going from low to high voltages. The values of the slope m of this variation for the different immersion times and for annealing temperatures varying from 50 °C to 200 °C are given in Table 4.

In the case of the Ag/McPS/Si diode, the value of m varies from 1.2 to 2.2 at low voltages, indicating a transition from ohmic conduction to quasi-space-charge-limited conduction (SCLC) by increasing temperature. At high voltages, m decreases from 2.93 to 1.78, remaining close to

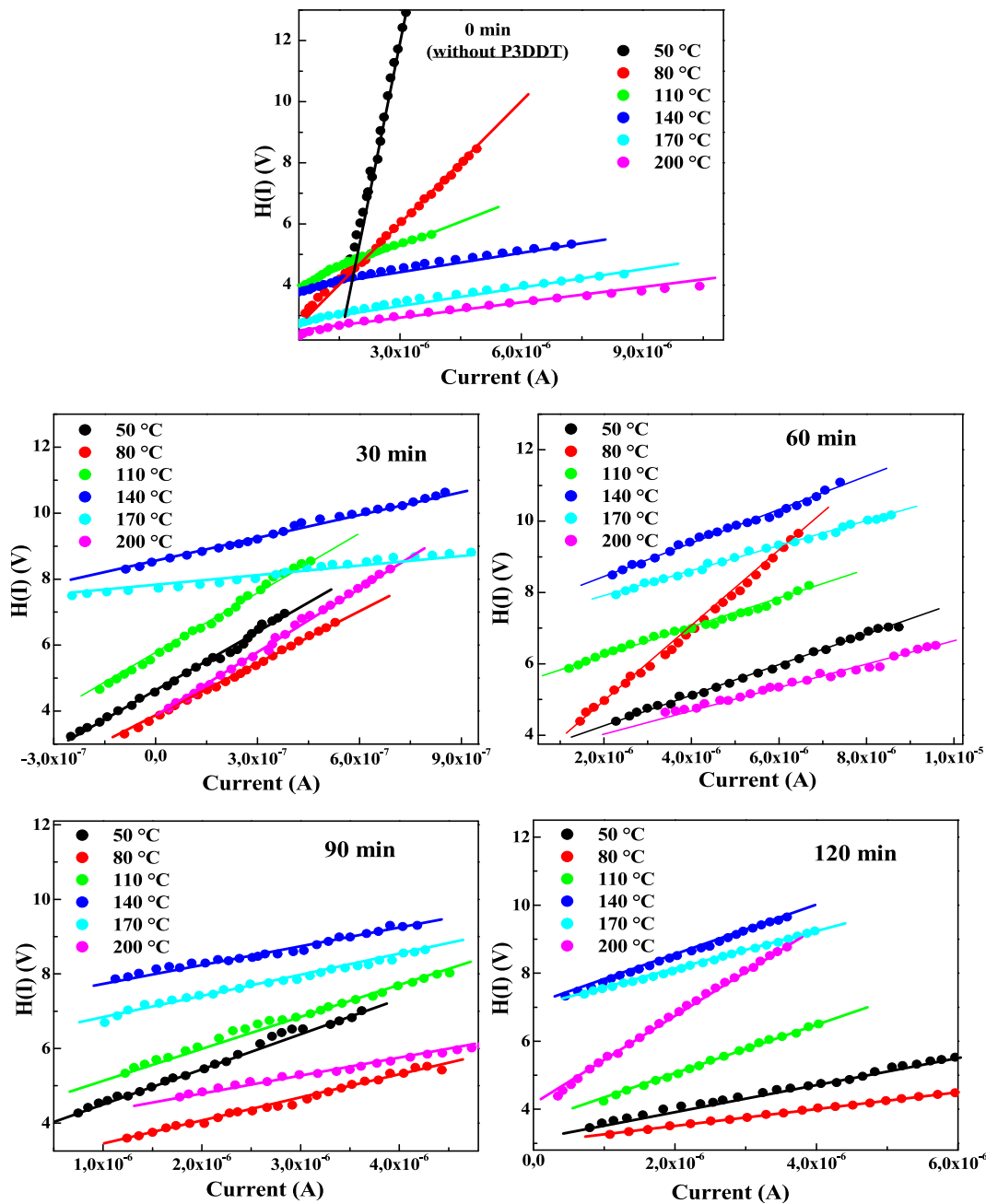


Fig. 11. The $H(I)$ plots of the Ag/P3DDT/MPS Schottky diodes for different immersion durations as a function of annealing temperature ranging from 50 °C to 200 °C.

2, which suggests that the SCLC mechanism remains dominant with barrier reorganization or traps deactivation induced by thermal effects.

The insertion of a P3DDT polymeric layer between the metal (Ag) and the semiconductor (McPS) significantly alters the charge transport mechanism, particularly under the influence of annealing temperature. In fact, for an immersion time of 30 min, we observe stable ohmic conduction at low voltage, while strong carrier injection with trapping, revealing a SCLC effect with deep traps, is evident at high voltages, reflected by very high m values (4–7). For medium immersion times (60 min and 90 min), moderate SCLC conduction is established for high polarizations. The polymer is likely well distributed, ensuring good injection without excessive trapping. For long immersion times of 120 min, a strong contribution of trap states is observed, even at low voltages, due to polymer accumulation or degradation after long immersion times. For low applied voltages, m varies from 1.4 to 3.1, which is very

sensitive to temperature, indicating mixed and complex transport behavior. The m slope varies greatly depending on the immersion time and annealing temperature. This allows identifying the optimal conditions for efficient transport with an immersion time of 90 min and moderate annealing (~140–170 °C), due to a homogeneous and low-trapping layer. The two Cheung equations (Eq. (2)) and (Eq. (4)) are used to determine the diode parameters after the annealing treatment.

A good fit is found for all I-V characteristics showing linearity for either $dV(I)/d\ln(I)$ or $H(I)$ as a function of I as shown in Fig. 10 and Fig. 11. Organic interlayer Schottky diodes often exhibit high ideality factors, reflecting complex transport influenced by the structural, electronic, and morphological properties of the organic layer. Thermal annealing plays an important role in interface optimization. Indeed, slopes decrease with increasing temperature for immersion times below 60 min, indicating improved junction quality (Fig. 10). The optimal

Table 5

Diode parameters of Ag/P3DDT/McPS diode for different immersion times after various annealing temperatures.

	A.T.	50 °C	80 °C	110 °C	140 °C	170 °C	200 °C
$t_{\text{imm}} = 0$ min	n	5.26	4.13	3	2.26	6.42	7.88
	R_s (Ω)	9277	2293	711	664	1883	2551
	ϕ_b (eV)	0.69	0.77	1.79	1.53	0.95	0.65
$t_{\text{imm}} = 30$ min	n	5.77	2.5	3.73	3.15	2.15	3.46
	R_s (Ω)	5860	5444	6460	2459	2890	6573
	ϕ_b (eV)	0.8	1.46	1.53	3	2.02	1.1
$t_{\text{imm}} = 60$ min	n	10.27	11.69	4.69	4.92	7.62	5.27
	R_s (Ω)	422	1033	386	464	349	304
	ϕ_b (eV)	0.33	0.24	1.17	1.63	1	0.67
$t_{\text{imm}} = 90$ min	n	8.69	1.15	4.5	2.11	4.92	9.18
	R_s (Ω)	955	623	798	434	536	903
	ϕ_b (eV)	0.41	2.43	1	2.14	1.65	0.64
$t_{\text{imm}} = 120$ min	n	10.58	10.23	7.88	13.38	10.04	10.92
	R_s (Ω)	621	552	737	743	565	1337
	ϕ_b (eV)	0.56	0.67	0.56	0.65	0.85	0.44

immersion time appears to be around 60 min, beyond which performance degrades or stagnates.

At low annealing temperatures, the slopes of the $H(I)$ variations corresponding to the Ag/McPS diodes are very steep, indicating a high series resistance (Fig. 10). As the temperature increases, a relative thermal improvement is observed, reflected by a decrease in the $H(I)$ slope. After the deposition of P3DDT on the McPS surface, the curves exhibit better linearity, and significantly lower slopes are observed for short immersion times, indicating a substantial reduction in series resistance. The diode performance improves across all temperatures, with the most notable enhancement observed between 110 °C and 170 °C. For long immersion times, the curves remain linear but exhibit steep slopes at high annealing temperatures (200 °C), indicating an increase in series resistance. This behavior can be attributed to an excessive amount of polymer, which may introduce additional barriers or interface defects that hinder charge transport.

The different parameters obtained following Cheung equations fits are given in Table 5.

Table 5 presents the diode parameters, including ideality factor n , series resistance R_s , and Schottky barrier height ϕ_b for Ag/P3DDT/McPS diodes under various immersion times (t_{imm}) and annealing temperatures (A.T.).

The ideality factor is very sensitive to the amount of polymer deposited on the McPS surface and also to the annealing temperature, a large variation of this parameter is found with minimum values of 1.15 for diode obtained at $t_{\text{imm}} = 90$ min and annealed at a temperature of 80 °C. For the Ag/McPS diode, n decreases sharply with temperature up to 140 °C (from 5.26 to 1.96 at 140 °C), and then increases again, indicating non-ideal to near-ideal behavior around 110–140 °C. The series resistance R_s decreases with increasing temperature up to 140 °C, then rises again, indicating an improvement of the Ag/McPS contact under moderate annealing conditions. ϕ_b peaks at 2.14 eV (140 °C), but the values are inconsistent. Generally, the non-monotonic variation of the Schottky barrier in diodes incorporating an intermediate layer of organic polymer results from several interfacial phenomena. Charge trapping and injection into the polymer create internal electric fields that modulate the barrier [49]. P3DDT can also passivate or introduce surface states, affecting Fermi level blocking and leading to variations in the potential barrier [50]. The effect of passivation was

demonstrated in our previous study by examining the optical properties of McPS/P3DDT nanocomposites. Furthermore, the thickness and distribution of the P3DDT polymer on the porous surface, as well as the chemical interactions that lead to the formation of intermediate energy levels, can thus modify the apparent barrier [51,52]. These last two effects are also very likely considering the morphological observations and chemical finding. These results show that annealing improves the diode even without P3DDT, but the behavior remains irregular, confirming poor interface control. Immersion for 30 min at an A.T. of 170 °C provides optimal diode performance, but performance is less consistent outside this condition. For 60 min, the diode provides low resistance and stable performance, although not minimal. Immersion times exceeding 90 min lead to excess polymer, which degrades diode properties due to impediments to charge transport or defective interfaces. Indeed, for t_{imm} of 120 min, n is very high at all temperatures, indicating extensive trap-assisted currents within the polymer and non-ideal diode behavior. The best overall diode performance is obtained at 30 and 60 min of immersion and at A.T. of 170 °C and 200 °C, resulting in the lowest series resistance and stable parameters. Therefore, longer immersion times degrade performance due to polymer over-accumulation, while short to moderate immersion with optimized annealing results in the most efficient and stable Schottky junction.

Optimized immersion and annealing conditions in Ag/P3DDT/McPS Schottky diodes lead to significant improvements in interface quality and charge transport, providing valuable insights for interface engineering in organic/inorganic Schottky structures. The obtained findings demonstrate that controlled polymer deposition and thermal treatment can effectively tailor diode parameters. In addition, P3DDT/McPS structures exhibit strong potential for optoelectronic applications owing to their attractive optical properties, as reported in our previous study [51].

4. Conclusion

This study provides a comprehensive assessment of the morphological, chemical, and electrical modifications induced by P3DDT deposition and subsequent thermal annealing in Ag/P3DDT/McPS Schottky diodes. AFM and FTIR analyses have demonstrated that a moderate immersion time (30–60 min) ensures adequate molecular organization of P3DDT, while longer durations promote over-accumulation of this polymer, leading to increased roughness. Controlled deposition of P3DDT onto the porous matrix is therefore essential for establishing a well-defined organic-inorganic interface. The transport mechanism evolves with the amount of polymer, shifting from ohmic to SCLC behavior. Before annealing, diodes subjected to moderate immersion times exhibit better barrier formation and reduced series resistance, thus confirming that polymer distribution critically influences charge injection and trap distribution. Annealing in the temperature range of 140–170 °C significantly improved the rectifying behavior, reduced the ideality factor, and lowered the series resistance. These enhancements are attributed to polymer chain reorganization and trap passivation at both the Ag/P3DDT and P3DDT/McPS interfaces. The best-performing devices were obtained for immersion duration of 30 min and annealed at 170 °C, achieving ideality factors of 2.15 and a potential barrier of 2.02 eV. These results reinforce the potential of P3DDT-based heterojunctions and organic Schottky diodes (OSDs) for their integration into high-performance optoelectronic, sensing, and energy conversion technologies.

CRedit authorship contribution statement

Ahmed Hemdani: Writing – original draft, Visualization, Methodology, Formal analysis. **Hasan Assaedi:** Writing – review & editing, Supervision. **Mehdi Rahmani:** Writing – review & editing, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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